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CLASS:P5

WEEK-5 C PROGRAMS

PROGRAM-1

CODE

```c
#include<stdio.h>

int main()
{
    float daily_expense[7],total_expense=0;
    int count=0;

    printf("Enter expense for 7 days:\n");

    for(int i=0 ; i< 7;i++)
    {
        printf("Day %d: ",i+1);
        scanf("%f",&daily_expense[i]);
        total_expense+=daily_expense[i];

        if(daily_expense[i]>1000)
            count+=1;
        else{}
    }

    printf("Total expense = %.2f\n",total_expense);
    printf("Number of days expense was above 1000 = %d\n",count);

    return 0;



}
```

OUTPUT

```
PS C:\Users\Chethan S\Documents\Week-5 submit> cd
Enter expense for 7 days:
Day 1: 450
Day 2: 390
Day 3: 690
Day 4: 550
Day 5: 950
Day 6: 2309
Day 7: 4508
Total expense = 9847.00
Number of days expense was above 1000 = 2
PS C:\Users\Chethan S\Documents\Week-5 submit>
```

PROGRAM-2

CODE

```c
#include<stdio.h>

int main()
{
    int n,odd=0,even=0;
    printf("Enter the number of elements: ");
    scanf("%d",&n);

    int array[n];

    printf("Enter %d integers:\n",n);
    for(int i=0;i<n;i++)
    {
        scanf("%d",&array[i]);

        if(array[i]%2==0)
            even+=1;
        else if(array[i]%2!=0)
            odd+=1;
        else {}
    }


    printf("Number of Even numbers = %d\n",even);
    printf("Number of Odd numbers = %d\n",odd);

    return 0;
}
```

OUTPUT

```
PS C:\Users\Chethan S\Documents\Week-5 submit> cd "c
Enter the number of elements: 10
Enter 10 integers:
1 2 3 4 5 6 7 8 9 10
Number of Even numbers = 5
Number of Odd numbers = 5
PS C:\Users\Chethan S\Documents\Week-5 submit>
```

PROGRAM-3

CODE

```c
#include<stdio.h>

int main()
{
    int electricity[6],units=0,count=0;
    float avg_consumption;

    printf("Enter electricity units consumed by 6 houses:\n");

    for(int i=0;i<6;i++)
    {
        printf("House %d: ",i+1);
        scanf("%d",&electricity[i]);

        units+=electricity[i];

        if(electricity[i]>300)
            count+=1;
        else{}

    }

    avg_consumption=(float)units/6;

    printf("Total units consumed = %d\n",units);
    printf("Average consumption = %.2f\n",avg_consumption);
    printf("Number of houses consuming more than 300 units = %d\n",count);

    return 0;
}
```

OUTPUT

```
PS C:\Users\Chethan S\Documents\Week-5 submit> cd "c:\Us
Enter electricity units consumed by 6 houses:
House 1: 350
House 2: 420
House 3: 250
House 4: 345
House 5: 454
House 6: 654
Total units consumed = 2473
Average consumption = 412.17
Number of houses consuming more than 300 units = 5
PS C:\Users\Chethan S\Documents\Week-5 submit>
```

PROGRAM-4

CODE

```c
#include<stdio.h>

int main()
{
    int seats[5][4],booked=0,available=0,total;

    printf("Enter seat availability(1=Booked,0=Available)");

    for(int i=0;i<5;i++)
    {printf("Row %d:\n",i+1);
    for(int j=0;j<4;j++)
        {
            printf("Seat %d: ",j+1);
            scanf("%d",&seats[i][j]);

            if(seats[i][j]==0)
                available+=1;
            else if(seats[i][j]==1)
                booked+=1;
            else{}
        }
    }
    printf("Seat Available Matrix:\n");
    for(int i=0;i<5;i++)
    {printf("\n");
        for(int j=0;j<4;j++)
        {
            printf("%d ",seats[i][j]);
        }
    }
    printf("\n");

    printf("Total Booked Seats = %d\n",booked);
    printf("Total Available Seats = %d\n",available);

    return 0;
}
```

OUTPUT

```
PS C:\Users\Chethan S\Documents\Week-5 submit> cd "c:\Use
Enter seat availability(1=Booked,0=Available)Row 1:
Seat 1: 1
Seat 2: 0
Seat 3: 1
Seat 4: 0
Row 2:
Seat 1: 1
Seat 2: 1
Seat 3: 0
Seat 4: 0
Row 3:
Seat 1: 0
Seat 2: 0
Seat 3: 1
Seat 4: 1
Row 4:
Seat 1: 1
Seat 2: 0
Seat 3: 0
Seat 4: 1
Row 5:
Seat 1: 0
Seat 2: 1
Seat 3: 0
Seat 4: 0
Seat Available Matrix:

1 0 1 0
1 1 0 0
0 0 1 1
1 0 0 1
0 1 0 0
Total Booked Seats = 9
Total Available Seats = 11
```

CODE

```c
#include<stdio.h>

int main()
{
    int matrix[3][3],row_sum=0,col_sum=0,diagonal_sum=0,sec_diagonal_sum=0;

    printf("Enter elements of 3X3 matrix:\n");
    for(int i=0;i<3;i++)
    {
        for(int j=0;j<3;j++)
        {
            printf("Element[%d][%d]:  ",i,j);
            scanf("%d",&matrix[i][j]);
        }
    }
    printf("THE MATRIX IS:\n");

    for(int i=0;i<3;i++)
    {printf("\n");
        for(int j=0;j<3;j++)
        {
            printf("%d ",matrix[i][j]);
        }
    }
    printf("\n");
    printf("Sum of each row:\n");
    for(int i=0;i<3;i++)
    {
     row_sum=0;
        for(int j=0;j<3;j++)
        {
            row_sum+=matrix[i][j];
        }
        printf("Row %d sum = %d\n",i+1,row_sum);
    }

    printf("\n");

    printf("Sum of each column:\n");
    for(int j=0;j<3;j++)
    {

     col_sum=0;
        for(int i=0;i<3;i++)
        {
            col_sum+=matrix[i][j];

        }
        printf("Row %d sum = %d\n",j+1,col_sum);
    }

    for(int i=0;i<3;i++)
        {
            diagonal_sum+=matrix[i][i];
            sec_diagonal_sum+=matrix[i][2-i];

        }
    printf("\n");

    printf("Sum of main diagonal elements = %d\n",diagonal_sum);
    printf("Sum of secondary diagonal elements = %d\n",sec_diagonal_sum);

    return 0;
```

PROGRAM-5

OUTPUT

```
PS C:\Users\Chethan S\Documents\Week-5 submit> cd
Enter elements of 3X3 matrix:
Element[0][0]:  1
Element[0][1]:  2
Element[0][2]:  3
Element[1][0]:  4
Element[1][1]:  5
Element[1][2]:  6
Element[2][0]:  7
Element[2][1]:  8
Element[2][2]:  9
THE MATRIX IS:

1 2 3
4 5 6
7 8 9
Sum of each row:
Row 1 sum = 6
Row 2 sum = 15
Row 3 sum = 24

Sum of each column:
Row 1 sum = 12
Row 2 sum = 15
Row 3 sum = 18

Sum of main diagonal elements = 15
Sum of secondary diagonal elements = 15
PS C:\Users\Chethan S\Documents\Week-5 submit>
```